

```
Microsoft Windows [Version 10.0.26200.9168]
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C:\Users\DELL>mongosh
Current Mongosh Log ID: 6a96876a3d49f69c4f57df72
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.9.2
Using MongoDB:      8.3.7
Using Mongosh:       2.9.2
mongosh 2.10.0 is available for download: https://www.mongodb.com/try/download/shell
For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

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The server generated these startup warnings when booting
2026-08-28T08:26:37.487+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----
```

```
mongosh mongodb://127.0.0.1:27017
test> use bookflow
switched to db bookflow
bookflow> db
bookflow
bookflow> db.book_metadata.insertMany([
  {
    title: "The Silent Algorithm",
    author: "R. Menon",
    genre: "Technology",
    publishedYear: 2022,
    formats: ["Hardcover", "Digital", "Audiobook"],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: "K. Rao",
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: "M1001",
        rating: 5,
        comment: "Excellent deep dive into digital systems design.",
        date: ISODate("2023-01-15")
      },
      {
        member_id: "M1042",
        rating: 4,
        comment: "Good but the Digital edition had formatting issues.",
        date: ISODate("2023-02-20")
      }
    ]
  },
  {
    title: "Ocean of Code",
    author: "A. Fernandes",
    genre: "Technology",
    publishedYear: 2021,
    formats: ["Digital", "Paperback"],
```

```
formats: ["Digital", "Paperback"],
    specs: {
      pages: 210,
      fileSize_MB: 2.1,
      drm: false
    },
    avgRating: 4.2,
    reviews: [
      {
        member_id: "M2001",
        rating: 4,
        comment: "Solid Digital read, clean layout.",
        date: ISODate("2022-05-10")
      },
      {
        member_id: "M2002",
        rating: 5,
        comment: "Best programming book this year.",
        date: ISODate("2022-06-01")
      }
    ]
  },
  {
    title: "Whispers of Yesterday",
    author: "S. Kapoor",
    genre: "Fiction",
    publishedYear: 2019,
    formats: ["Hardcover", "Paperback"],
    specs: {
      pages: 410,
      binding: "Sewn"
    },
    avgRating: 3.8,
    reviews: [
      {
        member_id: "M3001",
        rating: 3,
        comment: "Slow start but strong ending.",
        date: ISODate("2020-01-15")
      }
    ]
  }
])
```

```

mongosh mongodb://127.0.0.1:27017/
> use bookflow
> db.book_metadata.insertOne(
  {
    member_id: "M3002",
    rating: 4,
    comment: "Beautifully written prose.",
    date: ISODate("2020-02-11")
  }
)
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a9687ad3d49f69c4f57df73'),
    '1': ObjectId('6a9687ad3d49f69c4f57df74')
  }
}

```

```

> db.book_metadata.find()
{
  '_id': ObjectId('6a9687ad3d49f69c4f57df73'),
  'title': 'The Silent Algorithm',
  'author': 'R. Menon',
  'genre': 'Technology',
  'publishedYear': 2022,
  'formats': [ 'Hardcover', 'Digital', 'Audiobook' ],
  'specs': {
    'pages': 340,
    'fileSize_MB': 4.2,
    'drm': true,
    'narrator': 'K. Rao',
    'runtime_min': 610
  },
  'avgRating': 4.6,
  'reviews': [
    {
      'member_id': 'M1001',
      'rating': 5,
      'comment': 'Excellent deep dive into digital systems design.',
      'date': ISODate('2023-01-15T00:00:00.000Z')
    },
    {
      'member_id': 'M1042',
      'rating': 4,
      'comment': 'Good but the Digital edition had formatting issues.',
      'date': ISODate('2023-02-20T00:00:00.000Z')
    }
  ]
},
{
  '_id': ObjectId('6a9687ad3d49f69c4f57df74'),
  'title': 'Ocean of Code',
  'author': 'A. Fernandes',
  'genre': 'Technology',

```

```
mongosh mongodb://127.0.0.1:27000
> use bookflow
> db.books.insertOne(
  {
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M2002',
        rating: 5,
        comment: 'Best programming book this year.',
        date: ISODate('2022-06-01T00:00:00.000Z')
      }
    ]
  }
)
> db.books.insertOne(
  {
    _id: ObjectId('6a9687ad3d49f69c4f57df75'),
    title: 'Whispers of Yesterday',
    author: 'S. Kapoor',
    genre: 'Fiction',
    publishedYear: 2019,
    formats: [ 'Hardcover', 'Paperback' ],
    specs: { pages: 410, binding: 'Sewn' },
    avgRating: 3.8,
    reviews: [
      {
        member_id: 'M3001',
        rating: 3,
        comment: 'Slow start but strong ending.',
        date: ISODate('2020-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M3002',
        rating: 4,
        comment: 'Beautifully written prose.'
      }
    ]
  }
)
```

```
mongosh mongodb://127.0.0.1:27000
> use bookflow
> db.books.insertOne(
  {
    _id: ObjectId('6a9687ad3d49f69c4f57df76'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  }
)
> db.books.insertOne(
  {
    title: 'The Silent Algorithm',
    avgRating: 4.6
  }
)
> db.books.insertOne(
  {
    title: 'Ocean of Code',
    avgRating: 4.2
  }
)
> db.book_metadata.find(
  {
    avgRating: { $gt: 4 }
  }
)
[
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'Ocean of Code', avgRating: 4.2 },
  { title: 'Quantum Horizons', avgRating: 4.9 },
  { title: 'Whispers of Yesterday', avgRating: 3.8 }
]
```

```

mongosh mongodb://127.0.0.1:27017
bookflow> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
)
[
  {
    title: 'The Silent Algorithm', avgRating: 4.6,
    title: 'Ocean of Code', avgRating: 4.2,
    title: 'Quantum Horizons', avgRating: 4.9
  }
]
bookflow> db.book_metadata.find(
  { formats: { $in: ["Digital", "Audiobook"] } },
  { title: 1, formats: 1, _id: 0 }
)
[
  {
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  { title: 'Ocean of Code', formats: [ 'Digital', 'Paperback' ] },
  { title: 'Quantum Horizons', formats: [ 'Digital', 'Audiobook' ] }
]
bookflow> db.book_metadata.find(
  { avgRating: { $gt: 4 },
    formats: { $in: ["Digital"] }
  })
[
  {
    _id: ObjectId('6a9687ad3d49f69c4f57df73'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
  }
]

```

```

mongosh mongodb://127.0.0.1:27017
reviews: [
  {
    member_id: 'M1001',
    rating: 5,
    comment: 'Excellent deep dive into digital systems design.',
    date: ISODate('2023-01-15T00:00:00.000Z')
  },
  {
    member_id: 'M1042',
    rating: 4,
    comment: 'Good but the Digital edition had formatting issues.',
    date: ISODate('2023-02-20T00:00:00.000Z')
  }
],
{
  _id: ObjectId('6a9687ad3d49f69c4f57df74'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [ 'Digital', 'Paperback' ],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: ISODate('2022-05-10T00:00:00.000Z')
    },
    {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book this year.',
      date: ISODate('2022-06-01T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6a9687ad3d49f69c4f57df76'),

```

```
mongosh mongodb://127.0.0.1:27000
{
  rating: 5,
  comment: 'The Digital format made annotations so easy.',
  date: ISODate('2023-08-01T00:00:00.000Z')
},
{
  member_id: 'M002',
  rating: 5,
  comment: 'Mind-bending and well narrated.',
  date: ISODate('2023-09-14T00:00:00.000Z')
}
]
}
}
bookflow> db.book_metadata.find(
  {
    formats: { $regex: "Digital", $options: "i" },
    { title: 1, formats: 1, _id: 0 }
  }
)
[
  {
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  {
    title: 'Ocean of Code', formats: [ 'Digital', 'Paperback' ] },
  {
    title: 'Quantum Horizons', formats: [ 'Digital', 'Audiobook' ] }
]
bookflow> db.book_metadata.find(
  {
    "reviews.comment": { $regex: "Digital", $options: "i" },
    { title: 1, "reviews.$": 1, _id: 0 }
  }
)
[
  {
    title: 'The Silent Algorithm',
    reviews: [
      {
        member_id: 'M001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  },
]
```

```
mongosh mongodb://127.0.0.1:27000
{
  $project: {
    _id: 0,
    title: "$_id",
    author: 1,
    avgReviewRating: {
      $round: [ "$avgReviewRating", 2 ]
    },
    totalReviews: 1
  }
}
]
}
]
[
  {
    totalReviews: 2,
    author: 'R. Menon',
    title: 'Quantum Horizons',
    avgReviewRating: 5
  },
  {
    totalReviews: 2,
    author: 'R. Menon',
    title: 'The Silent Algorithm',
    avgReviewRating: 4.5
  },
  {
    totalReviews: 2,
    author: 'A. Fernandes',
    title: 'Ocean of Code',
    avgReviewRating: 4.5
  }
]
bookflow> db.book_metadata.createIndex({
  "reviews.comment": "text"
})
reviews.comment_text
bookflow> db.book_metadata.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  {
    v: 2,
  }
]
```

```

mongosh mongodb://127.0.0.1:27017
> use bookflow
bookflow> db.book_metadata.find(
  { $text: { $search: "Digital" } }
).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: '4FB85230',
    planCacheShapeHash: '4FB85230',
    planCacheKey: 'F34F3876',
    optimizationTimeMillis: 1,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nss: 'bookflow.book_metadata',
      indexPrefix: {},
      indexName: 'reviews.comment_text',
      parsedTextQuery: {
        terms: [ 'digit' ],
        negatedTerms: [],
        phrases: [],
        negatedPhrases: []
      },
      textIndexVersion: 3,
      inputStage: {
        stage: 'FETCH',
        nss: 'bookflow.book_metadata',

```

```

mongosh mongodb://127.0.0.1:27017
> use bookflow
bookflow> db.book_metadata.find(
  {
    "reviews.comment": {
      $regex: "Digital",
      $options: "i"
    }
  }
).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow.book_metadata',
    parsedQuery: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
    indexFilterSet: false,
    queryHash: '819E6067',
    planCacheShapeHash: '819E6067',
    planCacheKey: '29FFD58A',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
      nss: 'bookflow.book_metadata',
      direction: 'forward'
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 0,
    totalKeysExamined: 0,
    totalDocsExamined: 4,
    executionStages: {
      isCached: false,
      stage: 'COLLSCAN',

```

```
mongosh mongodb://127.0.0.1:27017/
bookflow> db.book_metadata.createIndex({
  formats: 1
})
formats_1
bookflow> db.book_metadata.find(
  { avgRating: { $gt: 4 } }
).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow.book_metadata',
    parsedQuery: { avgRating: { '$gt': 4 } },
    indexFilterSet: false,
    queryHash: 'E21FCB34',
    planCacheShapeHash: 'E21FCB34',
    planCacheKey: 'B596F181',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: { avgRating: { '$gt': 4 } },
      nss: 'bookflow.book_metadata',
      direction: 'forward'
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 0,
    totalKeysExamined: 0,
    totalDocsExamined: 4,
    executionStages: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: { avgRating: { '$gt': 4 } },
      nReturned: 3,
    }
  }
}
```